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Date

Inverse Functions & Transformation

Inverse of a Function:

⇒ If you have a funⁿ f that maps an element x from set X to an element y in a set Y , then the inverse funⁿ f^{-1} maps y back to x .

$$\therefore \text{If } f: X \rightarrow Y$$

$$\text{then } f^{-1}: Y \rightarrow X$$

For every $y \in Y$ there is a unique $x \in X$, such that $f(x) = y$

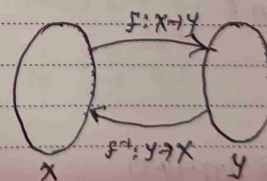
⇒ The inverse funⁿ f^{-1} satisfies the following condition:

$$\textcircled{1} \forall x \in X$$

$$f(f^{-1}(y)) = y$$

$$\textcircled{2} \forall y \in Y$$

$$f^{-1}(f(x)) = x$$



(हम है हैं)

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⇒ Identity funⁿ:

$$I_X: X \rightarrow X$$

$$\therefore \text{if } a \in X \\ \therefore I_X(a) = a$$

eg.,

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\therefore I \cdot v = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

⇒ Prop. of Identity funⁿ:

- ① Preservation → no alteration of element
- ② Linearity → I_X is a linear transformation
- ③ It be its own inverse.

⇒ A funⁿ f has an inverse iff it is bijective:

- ① one-to-one → injective
- ② onto → surjective

eg., $y = 2x + 3$

$$\therefore x = (y - 3) / 2$$

$$\therefore f^{-1}(y) = \frac{(y - 3)}{2}$$

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Appⁿ of Funⁿ and Inverse Function:

① Normalization & standardization

Feature scaling
with min max
normalization

we take every feature &
transformed into new feature
where $\mu = 0$ & $\sigma = 1$
↓
standard normal
distribution

Inverse of a Matrix:

$$\Rightarrow A = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$$

to find the inv. of A ,
 $T: \mathbb{R} \rightarrow \mathbb{R}$

say, $A \cdot x = y$

Now,

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$$

$$= \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$|A| = ad - bc$$

Note:

$|A| \neq 0 \rightarrow$ Inv. exists